1. For the questions 1-6 consider the following and choose the correct answer:

$v(t) = 200\cos(628t + 60^\circ)$ and $i(t) = -20\sin(628t - 130^\circ)$

The voltage leads the current in degrees by:

a) 100 b) 628 c) 314 d) _____

2) The current lags the voltage in radians by:

a) 1.57 b) 3.14 c) 1.74 d) _____

3) The voltage value (v) at t=5ms is:

a) 50 V b) -100 V c) -150 V d) _____

4) The phasor domain representation of i(t) is:

a) $20\angle -130$ V b) $40\angle -40$ V c) $20\angle -40$ V d) _____

5) The phasor domain representation of v(t) is:

a) $200\angle 60$ V b) $100\angle -60$ V c) $200\angle 130$ V d) _____

6) The frequency of the voltage in Hz is:

a) 50 b) 100 c) 314 d) _____

Problem 2. For the questions 7-10 consider the circuit shown in figure 1 and answer using nodal analysis:

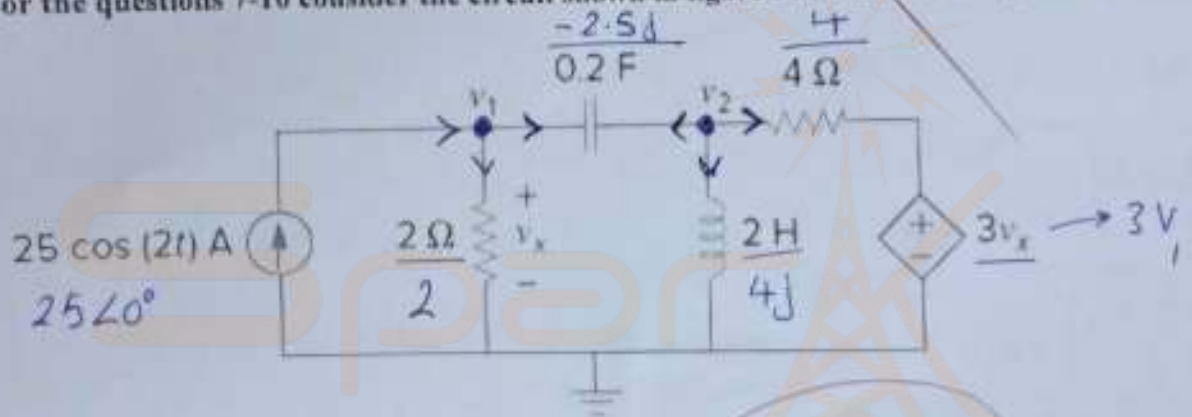


Figure 1

7) The voltage (Vx) is:

a) $28.31\angle 60$ V b) 67.6+ j10 V

c) $77.682\angle 50.19°$ V d) _____

8) The current through the inductor is:

a) $24\angle -90$ A b) 15+ j10 A

c) $20.64\angle -32.9°$ A d) _____

9) The voltage of node 2 with respect to node 1 is:

a) $12\angle 0$ V b) 5+ j10 V

c) $54.3\angle 55.6°$ V d) _____

10) write the required nodal equations (do not simplify):

① node (1) → $25\angle 0° = \frac{V_1}{2} + \frac{V_1 - V_2}{-j5}$

② node (2) → $\frac{V_2 - V_1}{-j5} + \frac{V_2}{4j} + \frac{V_2 - 3V_1}{4} = 0$

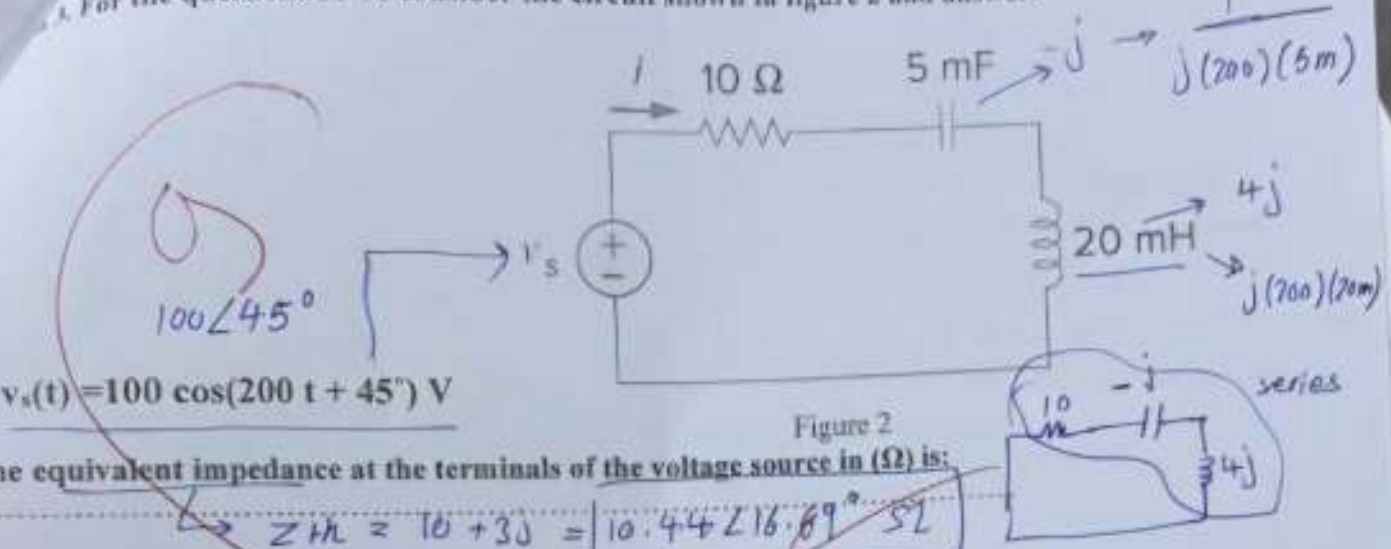
$\Sigma i_n = \Sigma i_{out}$

$V_x = V_1$

$\therefore 3V_x = 3V_1$

without simplify

...s. For the questions 11-16 consider the circuit shown in figure 2 and answer:



- $v_s(t) = 100\cos(200t + 45^\circ)$ V

Figure 2

11) The equivalent impedance at the terminals of the voltage source in (Ω) is:

$Z_{th} = 10 + 3j = 10.44\angle 16.69^\circ\ \Omega$

12) The admittance at the terminals of the voltage source in (S) is:

$Y = \frac{1}{Z} = \frac{1}{10+3j} = 0.0917 - 0.0275j$

13) The steady state current (i) in time domain is:

$I = \frac{V}{Z} = \frac{100\angle 45^\circ}{10+3j} = 9.57\angle 28.3 = 9.57\cos(200t + 28.3^\circ)$ A

14) The average power supplied by the source is:

$P_s = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{1}{2}(100)(9.57)\cos(45 - 28.3^\circ)$

$= 458.318$ W

15) The average power absorbed by the resistor is:

$V_R = (10)(9.57\angle 28.3) = 95.7\angle 28.3$

$P_r = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = \frac{1}{2}(95.7)(9.57)\cos(28.3^\circ - 28.3)$

$= 458$ W

16) The average power absorbed by the inductor and the capacitor is :

$P_c = 0$ W and $P_L = 0$ W

$\rightarrow \sum P_{supp} = \sum P_{absorb}$

458 W = 458 W ✓

Prob. 4: Consider the circuit shown in Figure 3 and answer questions 17-18.

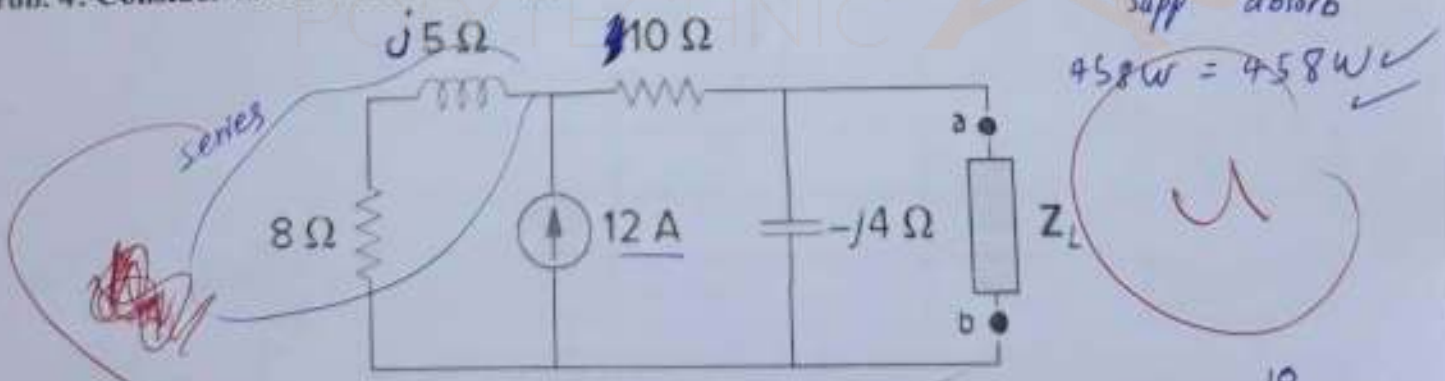


Figure 3

17.The Thevenin impedance with respect to terminals a-b is:

$Z_{th} = (8+5j+10) \parallel (-4j) = 0.886 - 4.04j$

18.The Thevenin voltage with respect to terminals a-b is:

$V_{Th} = (113.2\angle 32)\frac{-4j}{8+5j+10} \rightarrow V_{Th} = 25.11\angle -61.17^\circ$ V

by voltage divider

Good Luck

